

CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

# Chapter 4: Exploring Some Geometric Themes

Topic-wise short notes with solved examples

Two themes in geometry: fractals, which repeat the same pattern at smaller and smaller scales, and ways of visualising solids through nets, shortest paths, projections (front, top and side views) and isometric drawings.

## 1 Fractals

- A **fractal** is a self-similar shape: it shows the same pattern again and again at smaller and smaller scales.
- In nature: ferns, trees, clouds, coastlines, mountains and lightning.
- In art and architecture: the towers of temples such as the Kandariya Mahadev Temple at Khajuraho (about 1025 CE), patterns on Fulani wedding blankets of Nigeria, and prints by the artist M. C. Escher.
- A mathematical fractal is the result of repeating one geometric step for ever. Each stage of the repetition is called a **step**: Step 0, Step 1, Step 2, and so on.

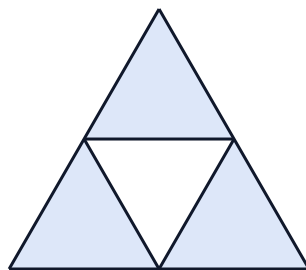
## 2 Sierpinski Carpet

- **Rule**: divide a square into 9 equal smaller squares and remove the central one. Repeat on each of the 8 squares that remain.
- Let  $R_n$  be the number of squares remaining and  $H_n$  the number of holes at Step  $n$ .
- Each remaining square produces 8 new squares:  $R_{n+1} = 8R_n$ , so  $R_n = 8^n$ .
- Each remaining square also produces one new hole:  $H_{n+1} = H_n + R_n$ , so  $H_n = 1 + 8 + 8^2 + \dots + 8^{n-1}$ .
- Each step keeps  $\frac{8}{9}$  of the area. Starting from area 1, the area at Step  $n$  is  $\left(\frac{8}{9}\right)^n$ .

| Step              | 0 | 1             | 2               | 3                 |
|-------------------|---|---------------|-----------------|-------------------|
| Squares remaining | 1 | 8             | 64              | 512               |
| Holes             | 0 | 1             | 9               | 73                |
| Area remaining    | 1 | $\frac{8}{9}$ | $\frac{64}{81}$ | $\frac{512}{729}$ |

## 3 Sierpinski Triangle

- **Rule**: join the mid-points of the sides of an equilateral triangle to get 4 congruent equilateral triangles, and remove the central one. Repeat on each of the 3 triangles that remain. It is also called the Sierpinski Gasket.



Step 1: the middle triangle is removed, three remain

- Triangles remaining at Step  $n$ :  $3^n$ . Holes:  $1 + 3 + 3^2 + \dots + 3^{n-1}$ . Area remaining:  $\left(\frac{3}{4}\right)^n$  of the original.
- Why the four triangles are congruent: each corner triangle has two sides equal to half a side with a  $60^\circ$  angle between them, so it is equilateral with side half the original. The middle triangle then also has three sides of that length.

**Example 3.1** How many triangles remain, and how many holes are there, at Step 4 of the Sierpinski Triangle?

**SOLUTION**

Triangles:  $3^4 = 81$ . Holes:  $1 + 3 + 9 + 27 = 40$ .

**81 triangles and 40 holes**

**Example 3.2** What fraction of the original area remains at Step 3 of the Sierpinski Triangle?

**SOLUTION**

$$\left(\frac{3}{4}\right)^3$$

$$\frac{27}{64}$$

## 4 Koch Snowflake

- **Rule:** start with an equilateral triangle. Divide each side into 3 equal parts, raise an equilateral triangle on the middle part and remove that middle part. Repeat on every side of the new shape. (Described by Helge von Koch in 1904.)
- Each side is replaced by 4 sides, each one-third as long.
- Number of sides at Step  $n$ :  $3 \times 4^n$ . Length of each side (starting from 1):  $\frac{1}{3^n}$ .
- Perimeter at Step  $n$ :  $3 \times \left(\frac{4}{3}\right)^n$ . The perimeter keeps growing without limit, although the shape stays inside a fixed region.

**Example 4.1** Find the number of sides and the perimeter of the Koch Snowflake at Step 2, starting from a triangle of side 1 unit.

SOLUTION

Sides:  $3 \times 4^2 = 48$ , each of length  $\frac{1}{9}$ .

**48 sides; perimeter  $\frac{48}{9} = 5\frac{1}{3}$  units**

## 5 Solids: faces, edges, vertices

- **Faces** are the flat surfaces of a solid. **Edges** are the line segments where faces meet. **Vertices** are the points where edges meet. A cuboid has 6 faces, 12 edges and 8 vertices.
- A **prism** has two congruent polygons as opposite faces, joined by parallelograms. It is named after the polygon: triangular prism, pentagonal prism, and so on.
- A **pyramid** has a polygon as base and a point (apex) joined to every vertex of the base. A triangular pyramid is a **tetrahedron**.
- An **octahedron** is two square pyramids joined at their bases: 8 triangular faces. A **dodecahedron** has 12 pentagonal faces.

| Solid                            | Faces   | Edges | Vertices |
|----------------------------------|---------|-------|----------|
| Prism on an $n$ -sided polygon   | $n + 2$ | $3n$  | $2n$     |
| Pyramid on an $n$ -sided polygon | $n + 1$ | $2n$  | $n + 1$  |
| Cube or cuboid                   | 6       | 12    | 8        |
| Tetrahedron                      | 4       | 6     | 4        |
| Octahedron                       | 8       | 12    | 6        |

- For all of these, Faces + Vertices – Edges = 2.

**Example 5.1** A prism has two 10-sided polygons as its ends. Find its faces, edges and vertices.

SOLUTION

Faces:  $10 + 2$ . Edges:  $3 \times 10$ . Vertices:  $2 \times 10$ .

**12 faces, 30 edges, 20 vertices**

**Example 5.2** A pyramid has a 10-sided base. Find its faces, edges and vertices.

SOLUTION

**11 faces, 20 edges, 11 vertices**

## 6 Profiles

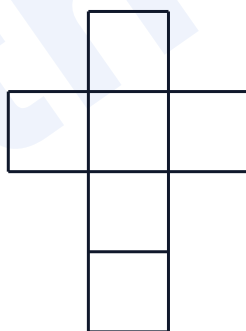
- What we see of a solid from one direction is its **profile**. The same solid can have very different profiles from different viewpoints.

| Solid                     | One profile               | Another profile       |
|---------------------------|---------------------------|-----------------------|
| Cylinder                  | Rectangle (from the side) | Circle (from the top) |
| Cone                      | Triangle (from the side)  | Circle (from the top) |
| Triangular prism          | Rectangle                 | Triangle              |
| Cone with its top cut off | Trapezium                 | Circle                |
| Pentagonal prism          | Pentagon                  | Rectangle             |

- A profile does not fix the solid: a sphere, a cylinder and a cone can all show a circle.

## 7 Nets

- A **net** is a flat shape that can be folded to make a solid. It is what you get when the solid is unfolded onto a plane.
- A **cube** has 11 different nets (nets that differ only by a turn or a flip count as the same). Each has 6 squares, but not every arrangement of 6 squares folds into a cube.



One of the 11 nets of a cube

- Cuboid**  $l \times b \times h$ : six rectangles in three matching pairs.
- Regular tetrahedron**: 4 equilateral triangles; it has 2 different nets. **Square pyramid**: a square with a triangle on each side.
- Cylinder**: two circles and a rectangle. The rectangle has one side equal to the height and the other equal to the circumference  $2\pi r$  of the circle.
- Cone**: a circle and a sector of a larger circle. The radius of the sector is the slant height of the cone.
- Octahedron**: 8 equilateral triangles; like the cube, it has 11 nets. A dodecahedron has 43,380 nets.
- A sphere has no net: a flat sheet cannot wrap a ball without wrinkles or gaps.

**Example 7.1** A cylinder has radius 7 cm and height 10 cm. What are the sides of the rectangle in its net? (Take  $\pi = \frac{22}{7}$ .)

SOLUTION

One side is the height. The other is the circumference  $2 \times \frac{22}{7} \times 7$ .

**10 cm and 44 cm**

**Example 7.2** Six squares are joined in a straight row. Is this a net of a cube?

SOLUTION

When folded, the row wraps round only 4 side faces and then overlaps. The top and bottom stay open.

**No**

## 8 Shortest paths on a cuboid

- On a plane the shortest path between two points is the straight segment joining them.
- A path on the surface of a cuboid becomes a path of the **same length** on its net. So unfold the cuboid, join the two points by a straight line and find its length, often with the Baudhāyana–Pythagoras theorem.
- The straight line must stay inside the net. Different ways of unfolding give different lengths, so **compare all the possible unfoldings** and take the smallest.

**Example 8.1** An ant sits at the centre of the front face of a cube of side 4 cm. A grain of sugar is at the centre of the top face. Find the shortest path along the surface.

SOLUTION

Unfold the top face upward. The ant is 2 cm below the shared edge and the sugar is 2 cm above it, on the same vertical line.

**4 cm**

**Example 8.2** A cuboid is 6 cm long, 3 cm wide and 2 cm high. An ant at a bottom corner wants to reach the opposite top corner. Find the shortest path along the surface.

SOLUTION

The three unfoldings give  $\sqrt{(6+3)^2 + 2^2} = \sqrt{85}$ ,  $\sqrt{(6+2)^2 + 3^2} = \sqrt{73}$  and  $\sqrt{(3+2)^2 + 6^2} = \sqrt{61}$ .

**$\sqrt{61}$  cm, about 7.8 cm**

## 9 Projections

- The **projection** of a point P on a plane is the point O of the plane such that OP is perpendicular to the plane. The projection of an object is made of the projections of all its points.
- The projection of a line segment is **never longer** than the segment. It has the same length only when the segment is parallel to the plane, and it shrinks to a point when the segment is perpendicular to the plane.
- Parallel lines project to parallel lines.** So the projection of a parallelogram is always a parallelogram (or a line segment).
- Different objects can have the same projection, so one projection is not enough. We use three perpendicular planes:

| View       | Plane            | Seen from |
|------------|------------------|-----------|
| Front view | Vertical plane   | The front |
| Top view   | Horizontal plane | Above     |
| Side view  | Side plane       | The side  |

- When sunlight falls perpendicular to a wall or floor, the **shadow** of an object is the same as its projection. A torch held close gives a larger, stretched shadow.

| Solid (standing upright) | Front view | Top view                  | Side view |
|--------------------------|------------|---------------------------|-----------|
| Cube                     | Square     | Square                    | Square    |
| Cuboid                   | Rectangle  | Rectangle                 | Rectangle |
| Cylinder                 | Rectangle  | Circle                    | Rectangle |
| Cone                     | Triangle   | Circle                    | Triangle  |
| Square pyramid           | Triangle   | Square with its diagonals | Triangle  |

**Example 9.1** Three identical cubes are placed in an L shape: two side by side on the ground and one on top of the left cube. Describe the front, top and side views.

**SOLUTION**

Front: two squares in a row with a third square above the left one (an L shape).

Top: two squares side by side. Side: two squares, one above the other.

**L shape; a 2 by 1 rectangle; a 1 by 2 rectangle**

**Example 9.2** A stick 10 cm long makes a projection 6 cm long on the floor. How much higher is one end than the other?

**SOLUTION**

The stick, its projection and the height difference form a right triangle:  $h^2 = 10^2 - 6^2 = 64$ .

**8 cm**

## 10 Isometric drawings

- In an **isometric projection** of a cube, all 12 edges appear with **equal length**. This happens when the cube is viewed along a line through opposite corners. Its outline is a regular hexagon.
- An **isometric grid** is made of dots (or lines) forming equilateral triangles. It has three directions: vertical for **height**, and two slanting directions for **length** and **depth**.
- To draw a solid: move along the three directions, counting units edge by edge. Parallel edges of the solid stay parallel in the drawing, and equal lengths stay equal.
- Draw faintly first, then darken only the edges that are visible.
- Four cubes joined face to face can be arranged flat in 5 ways (the "tetris" shapes) and in 3 more ways that are not flat, giving 8 solids in all.
- Some drawings on an isometric grid, such as the "impossible triangle", cannot be built as real solids. Each corner looks correct, but the whole does not fit together.

## 11 Common mistakes

- Thinking that the area of the Sierpinski Carpet at Step  $n$  is  $1 - \frac{n}{9}$ . Each step multiplies the area by  $\frac{8}{9}$ .
- Counting the base of a pyramid twice, or forgetting the two end faces of a prism.
- Assuming any six joined squares make a cube net.
- Taking the first unfolding as the shortest path without checking the others.
- Confusing the top view with the front view.
- Drawing an isometric cube with horizontal edges. Only the height direction is along a grid line that is vertical; the other two slant.